

HARIGOVIND P

AI/ML | Computer Vision | Python | Data Science

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PROFESSIONAL SUMMARY

AI/ML-focused Computer Science graduate with hands-on experience developing machine learning and computer vision applications using Python, PyTorch, TensorFlow, Scikit-learn, OpenCV, and Flask. Experienced in data preprocessing, model development, evaluation, and building AI solutions, including a YOLO-based road pothole detection and lane analysis system. Strong foundation in machine learning, deep learning, data analysis, and Python programming, with a strong interest in developing real-world AI solutions.

EDUCATION

LBS College of Engineering, Kasaragod

B.Tech in Computer Science and Engineering | 2022-2026

CGPA: 8.4/10

Relevant Coursework: Artificial Intelligence, Machine Learning, Data Structures & Algorithms, Object-Oriented Programming, DBMS, Operating Systems, Computer Networks

TECHNICAL SKILLS

Programming: Python, C

AI/ML Frameworks: PyTorch, Scikit-learn, OpenCV

Data Analytics: NumPy, Pandas, Matplotlib, Seaborn, Power BI

Database: MySQL

Tools: Git, GitHub, Linux, VS Code

PROJECTS

Road Pothole Detection & Smart Lane Analysis | Python, YOLO11, PyTorch, OpenCV, Flask

- Developed an AI-based road safety application using **YOLO11s** for pothole detection and computer vision techniques for lane detection and departure analysis.
- Trained the pothole detection model on a combined dataset of **8,007 images**, achieving **88.10% mAP@0.50** and **82.94% F1-score**.
- Implemented image preprocessing, augmentation, confidence filtering, and lane detection using **OpenCV**.
- Achieved approximately **22.9 FPS on GPU**, demonstrating real-time AI inference capability.

Breast Cancer Classification using Machine Learning | Python | Scikit-learn | Pandas | NumPy

- Developed and evaluated multiple classification models including **Logistic Regression, Decision Tree, Random Forest, SVM, and KNN**.
 - Applied preprocessing, feature selection, feature scaling, and hyperparameter optimization using **SelectKBest** and **GridSearchCV**.
 - Achieved **98.83% test accuracy using Random Forest** and evaluated model performance using confusion matrices and classification metrics.
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CERTIFICATIONS

- Python for Data Science – IBM Skills Network (2025)
 - TCS iON Career Edge – Young Professional (2025)
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LEADERSHIP

Volunteer Secretary – National Service Scheme | LBS College of Engineering (2024–25)

LANGUAGES

Malayalam (Native) • English (Professional) • Hindi (Working Proficiency)